

Original article

Multipurpose renewable energy resources based hybrid energy system for remote community in northern Ghana

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ABSTRACT

Many developing countries face the triple problems of unreliable supply of electricity, inadequate access to good drinking water supply and insecurity. The problem of electricity and water supply are more pronounced in remote areas due to lack of utility grid extension. This study examines the feasibility of using hybrid energy system consisting of solar PV and biodiesel generators in meeting the electricity and domestic water needs of a remote community in Ghana. The Hybrid Optimization Model for Electric Renewable (HOMER) software was used to carry out the analysis presented in this study. A daily electrical load of 104 kWh/day to meet the need of 100 households, small-scale businesses and a central water pumping system is estimated. Five different initial investment support scenarios, are considered: 100%, 75%, 50%, 25% and 0%. The results show a rather poor load profile (load factor of 12.26%) and a leveled cost of electricity (LCOE) of US\$0.76/kWh at full cost (100% cost). The LCOE declines to US\$0.20/kWh when the entire upfront cost of the project is grant-financed. Compared with end-user tariff of US\$0.10/kWh for lifeline customers in Ghana, beneficiaries of this PV-based mini-grid system will pay up to 200% more, even under full-grant finance conditions for maintenance of the system. Such interventions often replace inferior and health-threatening options such kerosene lamps, candles and in some cases outright darkness. Although the system is relatively expensive from a purely financial point of view, social equity considerations, together with innovative approaches encouraging daytime use of electricity for productive activities can help sustain such mini-grid solutions.

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Introduction

Access to basic energy and electricity services in Sub-Saharan Africa is a well-known developmental challenge. There is well over half a billion people in this region alone without access to electricity [1]. This is worrying due to the strong correlation between access to affordable and reliable energy services, and poverty reduction, improvement in health, growth in productivity, and economic and social development [2]. This knowledge is not lost in the mind of development partners, which were sure to include it in the Sustainable Development Goals. In fact, the Sustainable Energy for All (SE4All) initiative by the United Nations (UN) aims to make sustainable energy available globally by 2030 to over a billion people in the developing world [3–5]. Ghana is one of the first to develop a

national action plan for SE4All focusing specifically on access to modern energy services including clean cooking fuels [6].

As a developing country, electricity demand in Ghana was relatively low in the past, but there is now increasing demand resulting from economic growth, urbanization and other activities [7]. The Ghanaian economy has grown significantly over the past 20 years from a low-income to a lower-middle-income economy resulting in significant reduction in poverty levels by 28% from 1999 to 2013 [8–10] and with its attended urban migration and the rate of urbanization averaging nearly 4% per annum [11]. Without doubt, there is a number of challenges in the electricity sector in Ghana. The current installed capacity is not fully utilized due to a myriad of factors but mainly resulting from uncertainties in fuel supply constraints, and low rainfall patterns. The heavy dependence on hydro power is to blame for the negative impact of the low rainfall. More than 50% of the electricity is generated from hydropower plants [12]. In fact, variability and low rainfall patterns have significantly affected the power supply sufficiency and

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reliability in Ghana. This has compelled the power producers to shift their attention to expensive imported oil and gas-based electricity generation plants to supplement the shortfall. However, the power tariff regime, which is traditionally based on low cost hydropower makes cost recovery in the expensive oil-based generation plants impossible [7]. Furthermore, this adversely affects capacity expansion that is necessary to meet population growth [13], the lack of which is blamed for recent power shortages [14]. It is reported that Ghana needs additional annual capacity expansion of about 200 MW to meet the increasing demand [7]. In addition to the high cost of relying on oil and gas-based power plants to fill the supply gap is the problem of reliability and constraints of the oil and gas supply from Nigeria since Ghana does not produce sufficient oil and gas to meet its domestic needs [7,15].

Due to the increasing electricity supply challenges outlined above, there is the need to diversify the energy sources to include renewables. As a tropical country, Ghana has abundant solar energy resources that need to be harnessed to meet the energy needs of its people. Additionally, a conservative estimate suggests that up to a thousand square kilometers of land exists with moderate-to-excellent wind resource potential in the country to be exploited for electricity generation [12,13]. Despite the good renewable energy resource potential, less than 0.2% of Ghana's electricity is from variable renewable resources (that is, solar energy and wind energy) [7,16]. Access to electricity services in Ghana is associated with connection to the national electrical grid [6]. There is a huge disparity between access to electricity in the urban and rural populations. For example, the average national access to electricity in Ghana is about 71%, with that of the urban population at about 78% and the rural population at less than 30% [17]. Reported data revealed that by 2020, up to 3.2 million Ghanaians would not be connected to the national electricity grid [6]. Full national electricity access therefore depends on the extension of the grid or other appropriate alternatives to rural communities. For remote rural communities off-grid solutions using renewables such as solar, and wind can be the most cost effective [6]. Rural electrification is essential in reducing urban migration of youth from the rural communities with its attended negative consequences. Additionally, reported results indicate that income and urbanization significantly affect electricity consumption in Ghana with urbanization having the highest impact [18–20]. Furthermore, the per capita electricity consumption by the urban population is relatively higher in comparison with that of the rural population [18].

The Ghana government's national electrification program aims to provide electricity to all communities with minimum populations of 500 by 2020 [6,16]. It has therefore enacted the Renewable Energy Act mandating that 10% of the national electricity mix be renewables (excluding large-scale hydro) by 2020 [15]. Its overall policy objective focuses on expanding electrical power assets by investing in green energy technologies as well as energy conservation techniques [21]. As stated earlier Ghana has a reasonable amount of renewable energy resources including wind, solar, small hydro, and modern biomass that can be harnessed for electricity production [7,21]. Notwithstanding the wind and biomass resource potential, the renewable energy technologies to exploit them in a fashion that is sustainable and reliable for electricity production are not tested on a commercial scale. The challenges with employing stand-alone solar or wind energy sources is the fact that they can be unpredictable and unreliable [12], especially without expensive backup batteries. For example, a stand-alone solar energy system cannot provide sufficiently reliable electricity on cloudy days when there is no sunlight. Therefore the objective of the present study is to examine the economic analysis of a hybrid energy system consisting of solar PV and biodiesel generators in meeting the electricity and domestic water needs of a remote com-

munity in Ghana. The end goal is to provide new information and a range of options for policy makers and all stakeholders in the energy sector to make more informed decisions.

Material and methods

Description of study area and, energy and water needs

The Wa East district of the Upper West Region of Ghana was selected for this analysis. This was based on the fact that the 2010 national census revealed this region as having the highest out-immigration rate (28%) in the country and also had one of the lowest rates of electrification (30.9%) [22,23]. This regional average itself comprises a high of 77.4% for urban households and low of 18.9% for rural households (16,536 households) [23]. In Wa East district, only 9.2% of households (out of 10,768 households) had electricity as main source of lighting with 87.9% of the population relying on kerosene and flashlights for lighting [24]. The district has the lowest level of electrification in the Region. Fig. 1 shows the map of Ghana with enlarge size of Upper West region and Wa East district.

The proposed power plant for the selected community is designed to meet both primary and deferrable loads of the community. The primary loads are loads that must be met at particular times of the day. In this analysis, these loads comprise household and community level loads. Household level loads include lighting, television, radio, mobile phones and a fan. These loads were selected on the basis of the recommended minimum levels of electricity by the IEA and the UN SE4ALL initiative. In Ghana, low income households are regarded as those that consume up to 50 kWh of electricity in a month. This is the lifeline threshold. Street lighting loads are included as community level loads. Additionally, loads from small businesses such as cold stores and barbershops are included by providing 1400 W of power. To improve the load factor (defined as the ratio of the average load divided by the maximum (peak) load over a given period duration), it is assumed that these enterprises operate during the daytime. Table 1 presents a summary of the equipment/loads and the relevant assumptions in the load assessment.

Based on these loads and the assumed times and duration of use, the daily load profile is generated and shown in Fig. 2. This community would require a power plant that can produce 104 kWh of electricity per day with a peak load of about 34 kW. It is expected that there will be daily, monthly, seasonal variation in electricity consumption pattern in this community. To account for these variations, hourly and daily variations in the electrical loads are assumed to be 15% and 20%, respectively.

In addition to the primary electrical load, the community water supply system is also provided for, but as a deferrable load. A deferrable load is a load that must be met within some time period, but the exact timing is not important. A 1.3 kW dc pump is selected for this purpose. An average daily water requirement per person of 50 L [26] is assumed in this analysis. Based on an average national household size of 4.4 according to the 2010 census data of the Ghana Statistical Service [1], a daily community water usage need of 20 m³ (for 100 households) is estimated. A water storage capacity of 40 m³ is provided to enable water storage for an extra day. At full load, the 1.3 kW dc pump is able to pump water at a flowrate of 4 m³/h and will therefore run for 10 h in order to fill the storage tank. The energy storage capacity of the tank is 13 kWh, i.e. 1.3 kW * 40 m³/4 m³ h⁻¹. The daily community water requirement of 20 m³ could be met with the pump working at full load (1.3 kW) for 5 h (i.e. pumping at 4 m³/h). Therefore, the energy required to meet the daily water requirement is equivalent to 6.5 kWh. This is the daily deferrable load.

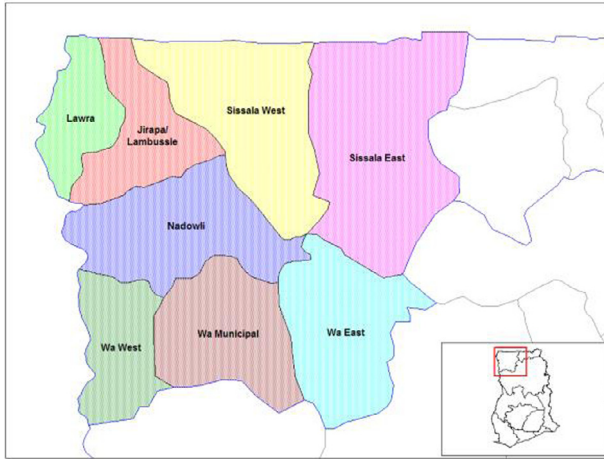


Fig. 1. Map of Ghana with enlarge size of Upper West region (proposed project site, Wa East, coloured in light blue) [25]. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 1

Equipment and loads for households and community (Primary Loads).

Load	Description	Time of use/Remarks
Lighting	3 light bulbs per household at 10 W each	0500–0600, 1800–2200
Radio	1 radio at 15 W	0600–0700, 1800–2100
Fan	1 fan at 30 W	1800–2100
Mobile phone	2 mobile phones at 6 W each	0700, 1800
Street lights	200 W for street lights	0000–0500, 1800–2300
Cold store	Cold Store 400 W daytime, 150 W during the night	0000–2300
Businesses	Small businesses (day time) – 1000 W	0800–1900

Solar energy resources

Due to lack of ground-based measured data, solar resource data was obtained using HOMER's online retrieval system linked to the NASA website and showed an average of 5.1 kWh/m²/day in July to 6.2 kWh/m²/day in March for the area, located at latitude 10°N, longitude 2.5°W. The annual variation is shown in Fig. 3. As expected, monthly variations in global solar radiation and hence, the monthly energy output from solar energy conversion systems would vary from one month to another. The annual averaged daily global solar radiation is 5.41 kWh/m²/day. The monthly clearness index, which is defined as the fraction of solar radiation at the top of the atmosphere that reaches a particular location on the earth surface varied between 0.494 (in August) and 0.639 (in December). The clearness index provides information on the level of

availability of solar radiation at the surface of the earth as well as changes in atmospheric conditions [27,28] and the weather condition at a site. Based on the range of values of the clearness index, the weather conditions in Wa East district can be classified as partly overcast.

Power plant configuration

The HOMER (Hybrid Optimization of Multiple Energy Resources) software, was used to conduct this analysis. HOMER was originally developed by the US National Renewable Energy Laboratory (NREL) and is currently owned by HOMER Energy LLC [29]. The proposed power plant configuration consist of solar PV array, energy storage facility (battery bank system), power converter and generator. The generator is expected to be powered by biofuel. A number of studies [30,31] have shown, that, *Jatropha curcas* could be grown in the northern part of the country, indeed the entire country is within the so called *jatropha curcas* belt [32]. The use of a biofuels instead of conventional diesel has the potential to create jobs in the local community and mitigate migration to the southern part of the country. The calorific value of the fuel bio-diesel is taken as 40 MJ/kg [33]. Figure 4 shows the configurations of the proposed system, while Table 2 presents the summary of the components, and the technical and financial input parameters.

Solar PV System

The solar PV system was modelled as fixed and tilted toward south at an angle of 15° to avoid settlement of dirt on the modules. The lifetime of the PV array (which is the same as the project life of the proposed hybrid energy systems) is taken to be 25 years. The derating factor, which compensates for efficiency reduction in the PV performance due to temperature, dust, and wiring losses was taken as 0.75 and the ground reflectance it is taken as 20% in this study. In addition, no tracking system is included in the PV system. For simulation analysis, four different PV array sizes: 10 kW, 20 kW, 30 kW and 40 kW; are considered. In addition, in order to investigate the effect of financial supports (or subsidies) on the economic performance of the proposed system, capital module subsidy levels of are simulated as 0% (no support), 25%, 50%, 75%, and 100% (fully supported). Technical specifications for selected module (Q-cell 255) for this installation are shown in Table 3. According to Denkyira [35], the cost of solar PV panel on the Ghanaian market ranges between \$1480/kW and \$3330/kW for module sizes of more than 150 W, with an average value of \$2230/kW. The relative high cost of PV modules in Ghana is due to lack of local PV production facilities in the country and hence, modules are generally imported (with transport, import duties and other related costs) from Asia, Europe or America. In this analysis, a PV cost of \$3000/kW is assumed.

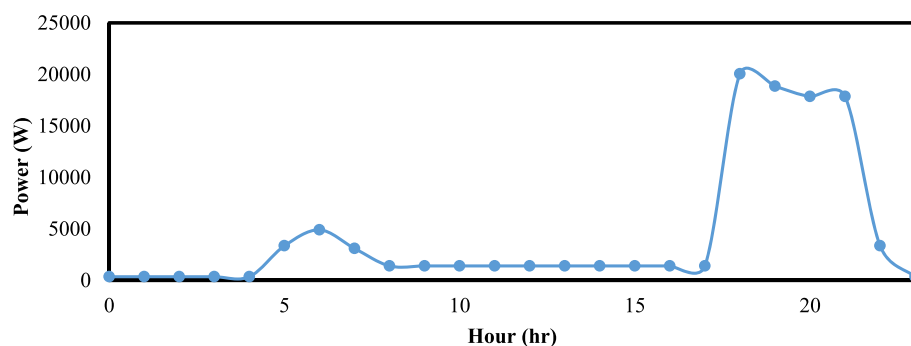


Fig. 2. Daily primary load profile.

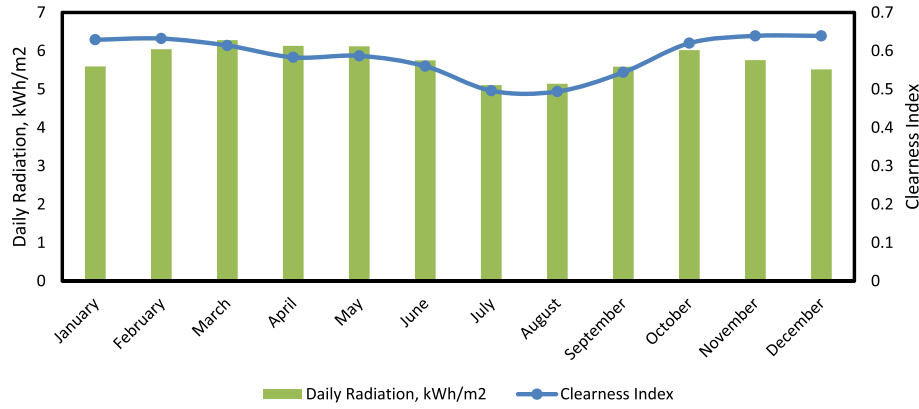


Fig. 3. Daily average monthly solar radiation (kWh/m²) and clearness index at Wa East District.

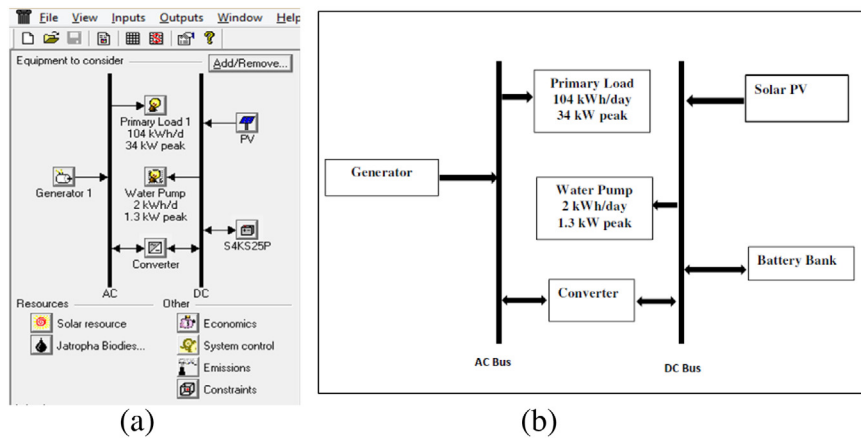


Fig. 4. Configuration of Solar PV-Biodiesel Community Hybrid Power System: (a) HOMER snapshot and (b) schematic diagram.

The output power from a PV system can then be calculated from the following equation [12]:

$$P_{PV} = P_{Rated} f \left(\frac{\overline{H}_T}{H_{T,STC}} \right) [1 + \alpha_p (T_c - T_{c,STC})], \quad (1)$$

where P_{Rated} is the rated capacity of the PV array or the power output under standard test conditions in kW; f is the PV derating factor; $\overline{H}_T$ is the solar radiation incident on the PV array (kWh/m²); $H_{T,STC}$ is the incident radiation at standard test conditions (1 kWh/m²); α_p is the temperature coefficient of power (%/°C) for the selected PV module; T_c is the PV cell temperature (°C) and $T_{c,STC}$ is the PV cell temperature under standard test conditions (25 °C). In a case where the effect of temperature on the PV array performance is neglected, α_p can be assumed to be zero and Eq. (1) reduces to:

$$P_{PV} = P_{Rated} f \left(\frac{\overline{H}_T}{H_{T,STC}} \right), \quad (2)$$

Generator

The generator system is used to supplement the power production by the solar PV energy conversion system and battery energy storage systems especially when the required electrical load is not fully provided by these systems. The fuel consumption from these generators can be estimated from the following expression [12,34]:

$$F_c = aP_{rated} + bP_{gen} \quad (3)$$

where F_c = generator fuel consumption (L), a = generator fuel curve intercept coefficient (L/h/kW), b = generator fuel curve slope (L/h/

kW), P_{rated} = generator rated capacity (kW) and P_{gen} = generator power output (kW). Two generators are used for this analysis. However, only one generator would be connected to the system and the other would act as a backup, especially during scheduled maintenance of the operating generator. Each generator is assumed to have running-time, total operation lifetime, of 15,000 h [37] and hence, the generators would have total combine running-time of 30,000 h. The operating lifetime of a generator can be determined from [29]:

$$R_{gene} = \frac{Q_{running-time}}{Q_{year}} \quad (4)$$

where R_{gene} = generator operational life (yr), $Q_{lifetime}$ = running-time of a generator (h), and Q_{thrp} = actual annual operating time (h/yr). For this analysis and based on initial simulation, the operation lifetime of a generator is found to about 9 years, after which, the generator would be replaced.

Battery and power converter

The battery is used to meet the electrical load during the non-availability of power from the energy generating systems (PV, wind turbine or diesel generator). In this study, the Surrrette 4KS25P model with a nominal capacity at 100-h rate of 1904 ampere hours (Ah) and cell voltage of 4 V is selected. The selected technical specifications for this battery are shown in Table 4. In order for the battery to produce more power, batteries are generally connected in series to form a battery string. In this analysis, a battery bank with 48 V is assumed and hence, 12 Surrrette 4KS25P batteries are connected series. Each battery string can produce about 91.2 kWh of

Table 2
Summary of Technical and Financial Input Data.

Parameter	Value	Remarks/Reference
Community	Wa East District	Lowest electrification rate in Region with highest migration rate [22,23]
No of households	100	Authors' assumption
Daily community load	104 kWh	Based on load assumptions
Water pumping (deferrable) load	2 kWh/day	Based on water needs assumptions [26]
Solar Resource	5.75 kWh/m ² /day	Lat. 10°N, long. 2.5°W
Solar PV capacity	20 kW	Sizes considered (kW): 10, 20, 30, 40
Solar PV installed cost	\$3000/kWp	Capital subsidy levels: 0%, 25%, 50%, 75%, 100%
Solar PV life	25 years	Typical PV warranty periods
Solar PV O&M	\$10/kWp-yr	Authors' assumption
Converter	30 kW	Sizes considered (kW): 10, 20, 30, 40
Converter cost	\$700/kW	Market data in Ghana [35]
Converter life	15 years	Typical
Converter O&M	\$10/year	Authors' assumption
Generator cost	\$1000/kW	2 alternating units. \$500/kW [35]
Biodiesel cost	\$1.1/L	[3] with calorific value of 40 MJ/kg [33]
Generator O&M	\$0.010/h	Authors' assumption
Generator operating hours	30,000	Initial cost accounts for 2 gensets, each with runtime of 15,000 h [37]
Min load ratio	30%	
Battery capacity	364.8 kWh	4 V Surrette 4KS25P batteries
Battery cost	\$200/kWh	Market data in Ghana [35]
Annual interest rate (real)	5%	
Project life	25 years	Authors
System fixed capital cost	\$50,000	To be used to develop a micro-distribution system, preparation of installation site, non-component related cost, etc
System fixed O&M	\$1500/yr	Labour and related cost

Table 3
Selected technical specification of a Q-cell 255 module [36].

Parameter (Units)	Value
Module Technology	Polycrystalline Silicon
Normal maximum power, P_{mpp} , (W)	255
Short-circuit current, I_{SC} , (A)	8.90
Open-circuit current, V_{OC} , (V)	37.83
Current at normal maximum power, I_{mpp} , (A)	8.37
Voltage at normal maximum power, V_{mpp} , (V)	30.77
Efficiency (nominal power) (%)	≥15.3
Temperature coefficients:	
P_{max} (%/K)	−0.42
I_{SC} (%/K)	+0.04
V_{OC} (%/K)	−0.30
Normal operating cell temperature, NOCT, (°C)	45 ± 3
Dimensions (mm)	1670 × 1000 × 35

electricity. Three different battery strings 2, 4, and 6 are considered. The battery life is determined by the software using the following equation [29]:

$$R_{batt} = \min\left(\frac{N_{batt} \cdot Q_{lifetime}}{Q_{thrpt}}, R_{battf}\right) \quad (5)$$

where R_{batt} = battery bank life (yr), N_{batt} = number of batteries in the battery bank, $Q_{lifetime}$ = lifetime throughput of a single battery (kWh), Q_{thrpt} = annual battery throughput (kWh/yr) and R_{battf} = battery float life (yr), taken as 12 years. Using Eq. (5), the battery life, in this analysis, is estimated to be 12 years. The capital and replacement cost of one battery is taken as \$200/kWh [35]. Using the energy balance approach, HOMER found that 4 strings (48 batteries) is an optimum solution for this energy system.

The proposed hybrid energy system comprises of both AC and DC systems, and hence a power converter system is required. A power converter maintains the flow of energy between the AC electrical load and DC components of the hybrid energy system. Since the hybrid energy system comprises both AC and DC systems, a power converter system is required. The lifetime of a unit is taken as 15 years with an efficiency of 90%.

Table 4
Technical specifications for Surrette 4KS25P battery [37].

Parameters	Value
Nominal capacity @ 100-h rate (Ah)	1904
Nominal voltage (V)	4
Round efficiency (%)	80
Minimum state of charge (%)	40
Float life (year)	12
Lifetime throughput (kWh)	10,569

System's performance metrics and definitions

Primary metrics used in this analysis are the Net Present Cost and the Levelized Cost of Electricity (LCOE). The NPC is the Present Value of all the costs incurred in the project's lifetime, minus the present value of all the revenue that it earns over its lifetime. It is obtained by [29]:

$$NPC = \frac{C_{ann,tot}}{CRF(i, R_{proj})} \quad (6)$$

where CRF is the Capital Recovery Factor, interest rate i (%) and project lifetime R_{proj} (yr). The Capital Recovery Factor (CRF) is, then calculated as:

$$CRF(i, N) = \frac{i(1+i)^N}{(1+i)^N - 1} \quad (7)$$

where i is the real interest rate and N is the number of years. The HOMER software computes the Levelized Cost of Energy (LCOE), the average cost per kWh of useful electrical energy produced by the system under consideration using [29]:

$$LCOE = \frac{C_{ann,tot} - c_{boiler}E_{thermal}}{E_{prim,AC} + E_{prim,DC} + E_{def} + E_{grid,sales}} \quad (8)$$

where $C_{ann,tot}$ is the total annualized cost (\$/yr); c_{boiler} is the boiler marginal cost (\$/kWh); $E_{thermal}$ is the total thermal load served (kWh/yr); $E_{prim,AC}$ is the AC primary load served (kWh/yr); $E_{prim,DC}$ is the DC primary load served (kWh/yr); E_{def} is the deferrable load

served (kWh/yr) and $E_{grid,sales}$ is the total grid sales (kWh/yr). Since there are no thermal loads considered, and no electricity sale to the grid, the expression for LCOE presented in Eq. (8) simplifies to:

$$LCOE = \frac{C_{ann,tot}}{E_{prim,AC} + E_{prim,DC} + E_{def}} \quad (9)$$

As shown in Table 2, various government/donor support scenarios are simulated to see the effect on viability in comparison with regular end-user tariffs charged by the public electricity distribution entity – the Northern Electricity Company (NEDCo). According to the Energy Commission of Ghana, the average end-user tariff for electricity increased from \$0.084/kWh in 2006 to \$0.147/kWh at the end of 2015 [38]. Using Eq. (10), a Cumulative Annual Growth Rate (CAGR) of 6.4% is estimated for electricity tariffs in Ghana.

$$CAGR = \left(\frac{T_{2015}}{T_{2006}} \right)^{1/(2015-2006)} - 1 \quad (10)$$

where, T_{2015} is the electricity tariff in 2015, T_{2006} is the electricity tariff in 2006. Eq. (11) is similarly used, with the CAGR of 6.4% to estimate future tariffs T_n in year n as:

$$T_n = T_{2015}(CAGR + 1)^{n-2015} \quad (11)$$

The Present Value of T_n is then estimated from:

$$PV(T_n) = \frac{T_{2015}(CAGR + 1)^{n-2015}}{(1 + d)^n} \quad (12)$$

$PV(T_n)$, in \$/kWh for each year is compared with the LCOE (also in \$/kWh) to determine the year in which it exceeds the LCOE. The Net Present Value of electricity purchases from grid is then computed at T_n is obtained as (Eq. (13)):

$$NPV(T_n) = Q_n \sum_{n=2015}^{N=25} T_{2015} \left(\frac{CAGR + 1}{1 + d} \right)^n \quad (13)$$

where the base year, 2015, has been set to zero and Q_n is the quantity of electricity that would be purchased from the grid if the community were connected. Q_n is the useful electricity consumption estimated by HOMER. NPV (T_n) is then compared with the NPC of this proposed project under various funding and support scenarios.

Results and discussion

The HOMER algorithm searched for the optimal solution among the various sizes considered for the defined load profile. This is accomplished by conducting hourly energy balance computations for various combinations of the equipment sizes specified. Feasible solutions were ranked according to Net Present Cost (NPC). The optimal system configuration suggested by HOMER is shown in Table 5 and comprises 30 kWp of Solar PV, 364.8 kWh of battery storage, 30 kW converter and 10 kW generator. In the reference scenario, i.e. at 100% cost and without any form of government or donor support, the investment cost, the net present cost and the LCOE are respectively estimated at: \$243,000, \$412,104 and \$0.76/kWh (Table 5). This LCOE, \$0.76/kWh, is over 400% higher than the average end-user tariff of \$0.15/kWh in Ghana [38]. Furthermore, when compared with tariff of \$0.10/kWh [39] for lifeline customers of the Northern Electricity Company (the public power distribution company in northern Ghana), the estimated LCOE for this proposed community hybrid power system is over 660% higher, if it were implemented without any aid.

The system' cost characteristics changes as the system configuration changes. For example, as shown in Fig. 5, a lower initial cost of \$213,000 is realized based on the configuration comprising 20 kW PV modules, 10 kW generator, 48 batteries and 30 kW converter. However, the NPC, LCOE and annual operating cost of this

Table 5

Technical and Financial Characteristics of Optimal System.

Parameter	Value
Solar PV	30 kW
Battery capacity	364.8 kWh
Converter	30 kW
Generator	10 kW
Annual Electricity Production	61,305 kWh
Excess Electricity	9787 kWh
Annual Electricity Consumed	38,471 kWh
Load factor	12.6%
Initial Capital	\$243,000
Net Present Cost (NPC)	\$ 412,104
LCOE	\$0.76/kWh

system are higher (\$428,036, \$0.79/kWh and \$15,257, respectively), when compared with optimal system solution. This is due to the increased use of the biofuel generator, from 1670 h to 2394 h.

On the other hand, electricity tariffs in Ghana have been increasing over the years at a rate of 6.4% per annual based on a 10-year Ghana Energy Commission data (2006–2015). With this rate of 6.4%, projected tariff growth during the life of the community hybrid power system is shown in Fig. 6. Tariffs evolve from \$0.15/kWh in year 0 to \$0.70/kWh in year 25, an increase of over 370%. However, since the LCOE takes into account the time value of money, the present value of tariffs is determined (shown also in Fig. 6) and shows a tariff growth of approximately 40%, from \$0.15/kWh to \$0.21/kWh on a present value basis.

Furthermore, the net present value of the cost of electricity consumed (38,471 kWh/yr) over the period, $NPV(T_n)$, if it was obtained from the national grid, is estimated based on Eq. (11) as \$169,046.94. This implies that communities on this proposed solar hybrid power system will pay over the life of the project, approximately 145% more compared with what would be paid if they were connected to the national grid. In addition, community dwellers using the proposed hybrid power system will end up paying at least 400% more in year 0 and 270% more in year 25, compared with communities that are connected to the national grid (Fig. 6).

It is seen, that, in the absence of any government or donor support, such a system is clearly not competitive with grid-based option, even when tariff escalation is taken into account. Therefore, various levels of government/donor support scenarios to such communities are explored to assess their impact on project viability metrics, mainly the LCOE (Fig. 7). The LCOE of \$0.76/kWh in the reference scenario, declines to \$0.20/kWh (and NPC of \$109,520) if the system is completely (100%) grant-financed and goes down further to \$0.16/kWh (and NPC of \$88,379) if the fixed O&M cost which covers items such as allowances for technicians is included in the grant financing arrangement.

In the suggested optimal system, PV and battery costs respectively comprise 37% (\$90,000) and 29.6% (\$72,000) of initial cost of \$243,000 (Fig. 8) and together constitute almost 67% of the total. On a life cycle basis, however, they together (modules and batteries) constitute 52.4% of the NPC, of which module cost contributes 22.9%.

The retail cost per Watt of PV modules in developed markets are quite well documented and reported by industry-led and research organizations such as SolarPower Europe (formerly EPIA¹), US Solar Energy Industries Association (SEIA), the US National Renewable Energy Laboratory (NREL) and the Fraunhofer Institute in Germany. On the other hand, markets in a developing region such as Ghana are not well developed and not well-studied for cost data and market

¹ European Photovoltaic Industry Association.

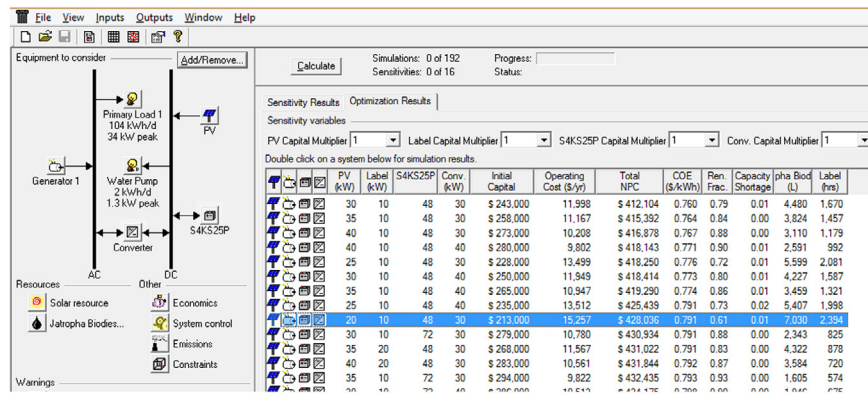


Fig. 5. Screenshot of feasible solutions and cost characteristics.

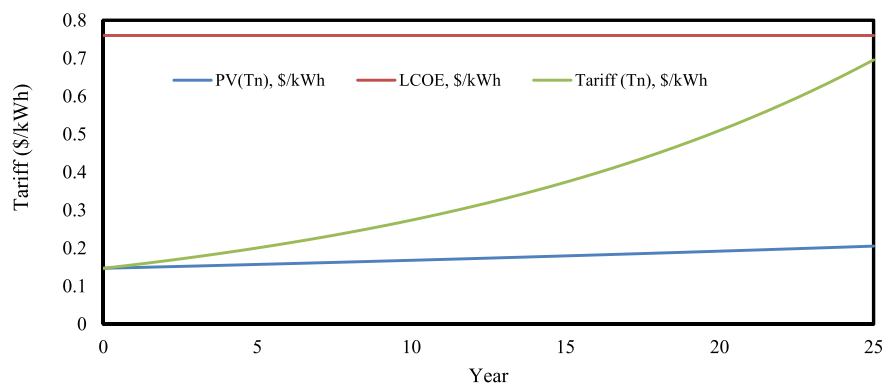


Fig. 6. LCOE of Proposed Community Solar-Biofuel Hybrid System vs Grid Tariff Evolution.

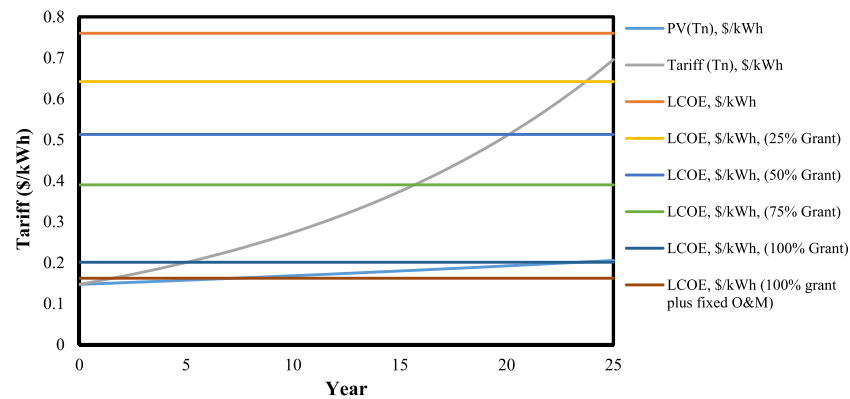


Fig. 7. Effect of Investment Support Level and Discount Rate on LCOE (\$/kWh).

trends. The low \$/W cost reported in markets such as Europe and North America are not necessarily present in relatively under-developed markets with small annual demand and inefficient supply and distribution channels. Additionally, megawatt scale projects (which are usually reported) tend to have cost characteristics that are quite different from small projects of the order of a few kilowatts.

The International Renewable Energy Agency (IRENA) 2016 report [40] on solar PV cost and markets in Africa assessed the cost of PV modules in off-grid mini-grid applications with batteries between the years 2011 and 2015. The report found the cost of modules to be in the range of \$0.80/W and \$2.80/W for seven projects. One project had an unusually high module cost component of \$6.30/W. On average, the PV modules constituted 26% of installed cost of the projects assessed.

Authors' personal communications with experts at Ghana's Energy Commission and some local retailers indicate that \$1000/kW were available on the local market at the close of 2016, if transportation and installation cost were excluded. This figure of \$1000/kW is lower than those reported by Denkyira [35]. Nevertheless, even in this optimistic cost scenario, an estimate of \$1500/kW appears more realistic in order to account for delivery, installation and other costs. In view of the rapid dynamics of the PV market and the declining cost of PV modules, the effect of module cost reduction (down from the \$3000/kW used in this study) on LCOE is analyzed and visualized in Fig. 9. At \$2000 (a multiplier of 0.67) and \$1000 (a multiplier of 0.33) per kW of PV module cost, the LCOE of the optimal system declines from \$0.76/kWh to \$0.70/kWh and \$0.65/kWh, respectively. Further electricity cost reductions

are realized if in addition to PV module cost, batteries costs also decline. Examples of similar studies conducted by various authors are shown in Table 6. The LCOE ranges from \$0.37/kWh [41] to \$1/kWh based on studies in Kenya [44].

Key observation from these studies is the close matching between solar resource availability and the load. The work by [43] for telecommunications base station in Doka-Sharia nature reserve in Nigeria particularly shows a good solar resource-load correlation. This results in a good load factor. Battery cost of \$174 as used by [41] and [43] for a 6 V, 360 Ah battery works out to \$80.56/kWh, which is remarkably low, compared with \$300/kWh by [44] and €300/kWh by [45] in the Kenyan study for German Development Bank KfW. Similarly, the cost of PV modules also showed a wide spread, from a low of \$600/kW for the system analyzed for the coastal regions of Bangladesh, to a high of \$4000 for the system in Thlatlaganya, South Africa.

An important observation made by [44] is the low load factor in such rural electricity projects, which contributes to high cost of electricity. In the present study, the low load factor contributes to a high LCOE. This is because the denominator in Eq. (7) (energy actually used) becomes smaller, resulting in a higher LCOE. The low load factor (as previously shown in Fig. 2) is mainly due to idle capacity and the low usage during the day-time as a result of such rural communities mainly being farmers, who spend most of the day on the farm and shift electricity usage to the evenings. Improvement in the load factor could be realized through investments in public service, small enterprises and other income-generating ventures that use power during the day and reduces the idle hours of the installed equipment.

The LCOE comparisons, as done, might not reveal the true extent of the benefits of such an intervention, since in reality, it is not displacing grid electricity, but rather, inferior, low-quality and often polluting options such as kerosene and flashlights [24]. Such investments may also be viewed as investment in health, productivity and jobs, and with the potential to mitigate migration from the communities. On rural electrification, the Electricity Company of Ghana, in its 2012 Annual Report [47] raised concerns about the low levels of return on its assets in rural communities and again in its 2013 tariff proposal to the Public Utilities Regulatory Commission (PURC) [48]. This was in view of low load factors and relatively low tariffs, since many tend to be on life-line tariffs – consuming up to 50 kWh/month. Yet, it is widely acknowledged, that, rural electrification may hardly be justifiable if viewed from the narrow perspective of financial and economic indicators. Indeed, the Government of Ghana has re-affirmed its commitment to pursuing universal access to electricity by 2020 [49], 10 years ahead of the 2030 target adopted by the UN under the SE4ALL initiative.

Conclusions

Lack of opportunities and essential services such as electricity and water are major contributors to migration, and in the case of Ghana, rural-urban migration. In this paper, we have conducted an analysis on the economic viability of using solar PV and biodiesel hybrid power system to supply to homes, enterprises and water in a community in the Upper West Region of Ghana without access to electricity. The cost of deploying this system of electricity service to a community of 100 households is estimated at \$243,000 with an LCOE of \$0.76/kWh. The LCOE of \$0.76/kWh is over 400% higher than the average end-user tariff of 0.15/kWh in Ghana. Capital investment support of 100%, however, brings the LCOE down to \$0.20/kWh and yields a superior net present cost of \$109,520 compared with \$169,046.94 for a grid-based option over the life of the project. This reinforces the key role of government and develop-

ment partners in providing communities with green and low-carbon options in electrification programmes. This role is particularly important in view of the characteristic of renewable energy technologies such as solar and wind, which require high upfront costs.

The regular financial metrics, such as LCOE and NPC are hardly able to reveal the true extent of the benefits of such an intervention. These interventions often displace inferior, low-quality and polluting options such as kerosene and locally made lamps, which pose health hazards to users. Such investments may rightly be viewed as investment in health, productivity and jobs, and with the potential to stem rural urban migration from such communities.

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